



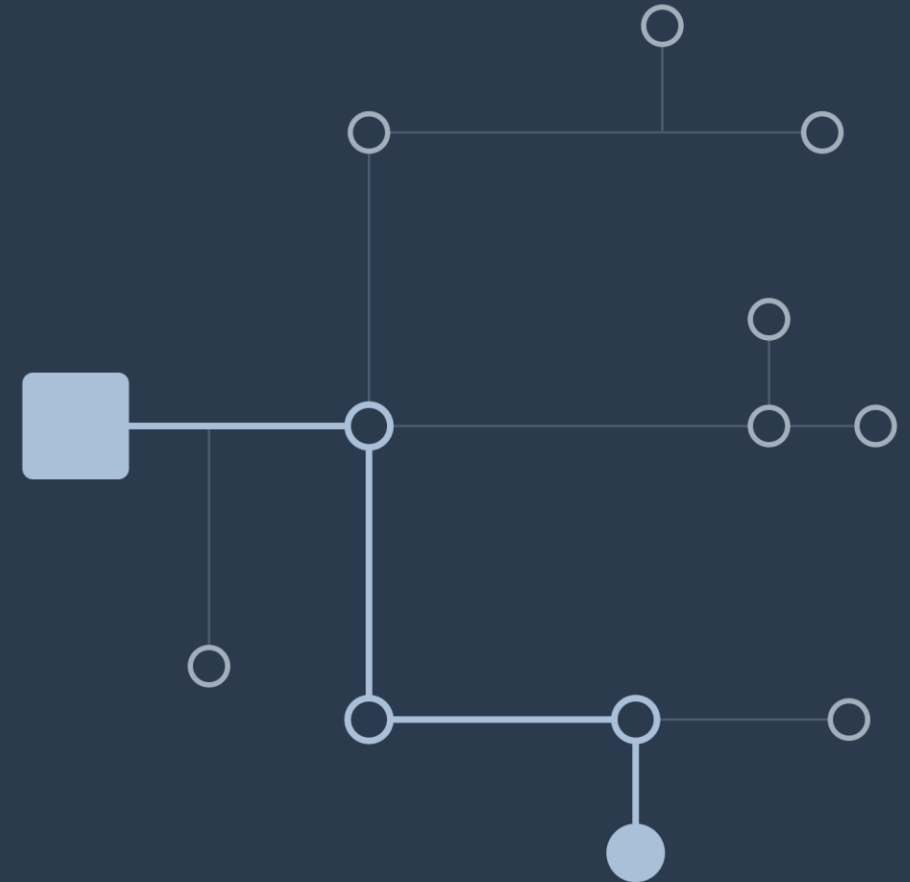
POLITÉCNICA

Grid Integration with Artificial Intelligence

A spatial AI copilot that lets a control room ask the network a question and get a grounded, auditable answer.

AI configured Agentic Grid System (A-AGS)

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THE PROBLEM

One outage question, five systems

SCADA

Live telemetry,
alarms and breaker
states

ADMS

Network model,
switching and load
flow

OMS

Customer calls,
outage records and
crews

GIS

Assets, locations,
connectivity and
ICPs

Documents

Procedures,
standards and past
incidents

Operators connect these by hand, under time pressure. The context lives in **people's heads**, not in the systems.



POLITÉCNICA

THE INSIGHT

The GIS network model is the context an AI agent needs before a control room can trust it.

Without topology, an LLM guesses. With a connectivity graph built from GIS and ADMS, every answer traces back to a feeder, a device and a source record.

THE PROPOSAL

Three services, one spatial foundation

01

Outage Triage Copilot

Correlates SCADA alarms and OMS calls with GIS traces to propose the probable fault section, affected ICPs and nearest crew.

Faster first assessment

02

Network Model Integrity Agent

Compares GIS edits against the ADMS model, flags connectivity and attribute errors, and drafts fixes for engineers to review.

A model operators trust

03

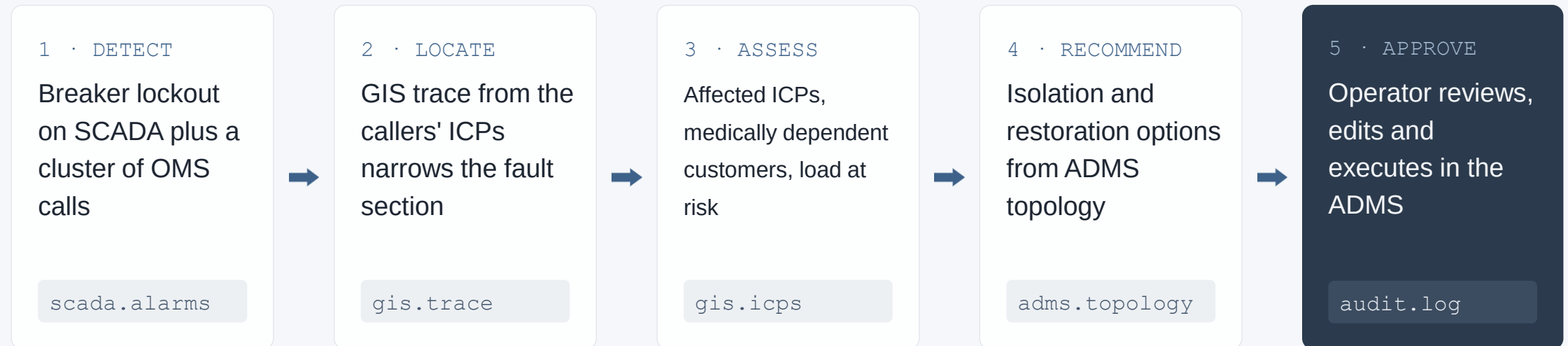
Switching and Field Assistant

Answers plain-language questions over the live network and drafts switching plans from topology, for operator approval.

Expert knowledge on demand

Shared foundation: CIM network graph from GIS + ADMS · RAG over procedures · MCP tool layer · human approval for every action

Outage triage in five agent steps



RAG at every step

Similar past incidents and the relevant operating procedure are retrieved and cited alongside each recommendation.

The agent proposes, the operator decides

GridSense never operates a device. Step 5 is always a person.

Keeping GIS and ADMS in agreement

GIS is where the network is drawn. ADMS is where it is operated. When the two drift apart, operators stop trusting the model.

Check	What the agent does
Connectivity	Finds floating conductors, unintended loops and phase mismatches in the traced graph
Attributes	Flags missing ratings, wrong voltage levels and stale normal-open states
Change impact	Traces each GIS edit to the feeders and ADMS objects it will change before export
Spatial sanity	Checks asset positions against aerial imagery and LiDAR where available
Output	A draft correction ticket with evidence: map extract, trace path and source records

ARCHITECTURE

A layered architecture built for OT

L5 Experience	Map-based copilot UI	Operator approval queue	Engineer review tickets	Teams and web access
L4 Agents	Supervisor agent	Triage, integrity, switching	Policy guardrails	Evaluation harness
L3 MCP tools	GIS server	ADMS and OMS servers	SCADA read server	Knowledge (RAG) server
L2 Context	CIM network graph	Vector index of procedures	Time-series cache	Audit store
L1 OT DMZ	Read-only replicas	CIM / CGMES exports	OPC UA, ICCP, REST	One-way data gateway
L0 Systems	SCADA and historian	ADMS	OMS and CIS	GIS

Security spine

IEC 62443 zones and conduits

Identity and role-based access per tool

LLM gateway with redaction

Every prompt and tool call audited

NZ-hosted or on-prem models

Each system becomes a governed toolset

MCP server	Example tools	Access
GIS	trace_upstream, trace_downstream, icps_on_section, assets_near	Read
ADMS	network_state, switch_status, draft_switching_plan	Read, draft only
OMS	active_outages, call_clusters, crew_locations	Read
SCADA	recent_alarms, point_history	Read via DMZ only
Knowledge	search_procedures, similar_incidents	Read, cited
Actions	create_ticket, submit_for_approval	Write, human-gated

Typed, permissioned and logged. Any LLM or agent framework can use them, so there is no lock-in to one model vendor, and each server is reusable across utilities.

Built for critical infrastructure from day one

Read-only by default

Agents read replicas in the OT DMZ.
There is no write path to SCADA.

Human approval

Every action is a draft until a qualified operator approves it.

IEC 62443 zoning

Components placed by Purdue level, with few and explicit conduits.

Grounded answers

Each response cites the trace, record or procedure it relied on.

Full audit trail

Prompts, tool calls and decisions logged for review and incident inquiry.

Data sovereignty

On-prem or NZ-hosted deployment, with the model choice kept open.

DELIVERY PLAN

A 20-week pilot on one network area

30 Days

Discover

Choose the use case and area with a client, complete security review, agree data access in the DMZ.

30 Days

Build the foundation

GIS-to-CIM graph, read-only MCP servers, RAG over procedures and past incidents.

45 Days

Shadow mode

Agents run on replayed and live events. Outputs are compared with what operators actually did.

45 Days

Evaluate

Control room trial, measured results and a go or no-go decision for production.

Deliverables: a working product, an evaluation report and a production roadmap, all reusable for the next utility.

SUCCESS MEASURES

How we will know it works

Measure	How it is measured	Target
Time to first assessment	Agent versus operator baseline on replayed outages	faster
Fault section accuracy	Proposed section versus confirmed fault location	%
Model issues found	GIS and ADMS discrepancies confirmed by engineers	per feeder
Grounding rate	Share of answers with a verifiable citation	%
Operator trust	Structured feedback after each shift	/ 5

Targets are set with the client in week 2, from their own baseline, so the result is credible to their board and regulator.

BUSINESS VALUE

From pilot to product

NOW

Pilot

One client, one network area, triage or model integrity. Proves value and builds the reference case.

NEXT

Product

Reusable MCP connectors per ADMS and GIS platform, packaged with implementation services.

LATER

Platform

GIS-based DER hosting capacity insight, and optimisation of restoration and crew scheduling on the same graph.

The aim: turn OT engineering know-how into a **repeatable product** that every new deployment makes stronger.